Benchmark Report

Autenticacao Pos-Quantica em Aplicacoes Web

ML-DSA-44 / Kyber512 vs RSA-2048 / RS256

3 execucoes independentes | N=100 iteracoes/run | 6000 amostras totais
Ambiente: Apple Silicon (ARM64), macOS, Python 3.13
liboqs 0.15.0 + liboqs-python 0.14.1

Gerado em: 2026-04-12

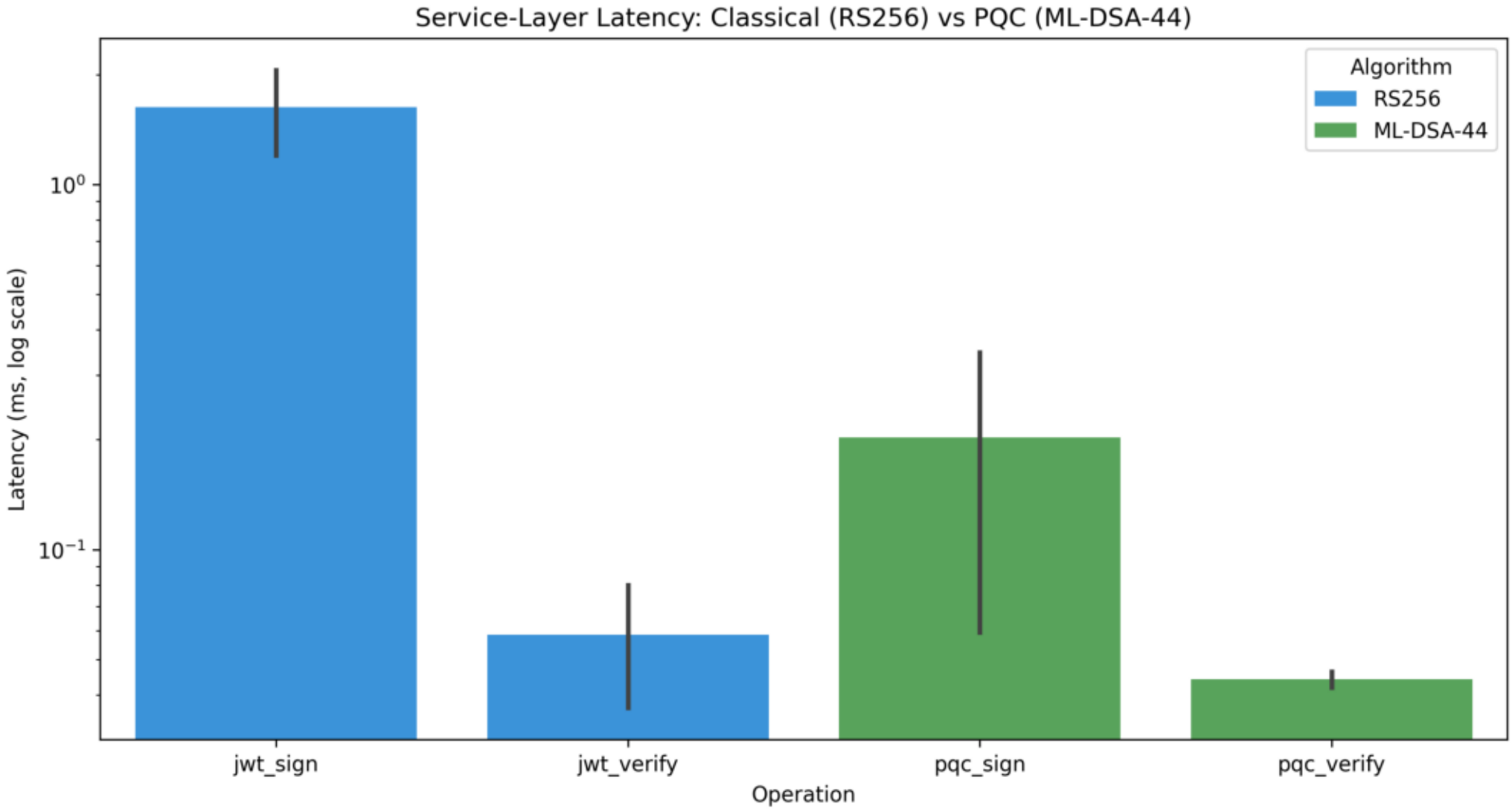
Reprodutibilidade Inter-Run (3 execucoes)

Operacao	Algoritmo	N Runs	Grand Mean (ms)	Inter-Run StDev (ms)	CV (%)
hybrid_sign_classical	RS256	3	1.6860	0.1051	6.24%
hybrid_sign_pqc	ML-DSA-44	3	0.2042	0.0172	8.41%
hybrid_verify_classical	RS256	3	0.0555	0.0005	0.92%
hybrid_verify_pqc	ML-DSA-44	3	0.0455	0.0003	0.67%
jwt_sign	RS256	3	1.6293	0.0720	4.42%
jwt_verify	RS256	3	0.0584	0.0053	9.01%
kem_decapsulate	Kyber512	3	0.0166	0.0001	0.90%
kem_encapsulate	Kyber512	3	0.0188	0.0002	0.98%
kem_keygen	Kyber512	3	0.0168	0.0001	0.33%
pqc_sign	ML-DSA-44	3	0.2031	0.0252	12.41%
pqc_verify	ML-DSA-44	3	0.0442	0.0006	1.34%
raw_kyber_decapsulate	Kyber512	3	0.0160	0.0001	0.79%
raw_kyber_encapsulate	Kyber512	3	0.0183	0.0003	1.55%
raw_kyber_keygen	Kyber512	3	0.0184	0.0004	2.23%
raw_mldsa_keygen	ML-DSA-44	3	0.0491	0.0027	5.48%
raw_mldsa_sign	ML-DSA-44	3	0.1061	0.0049	4.63%
raw_mldsa_verify	ML-DSA-44	3	0.0441	0.0007	1.60%
raw_rsa_keygen	RSA-2048	3	108.3329	3.9549	3.65%
raw_rsa_sign	RSA-2048	3	89.3226	0.8598	0.96%
raw_rsa_verify	RSA-2048	3	0.1131	0.0024	2.15%

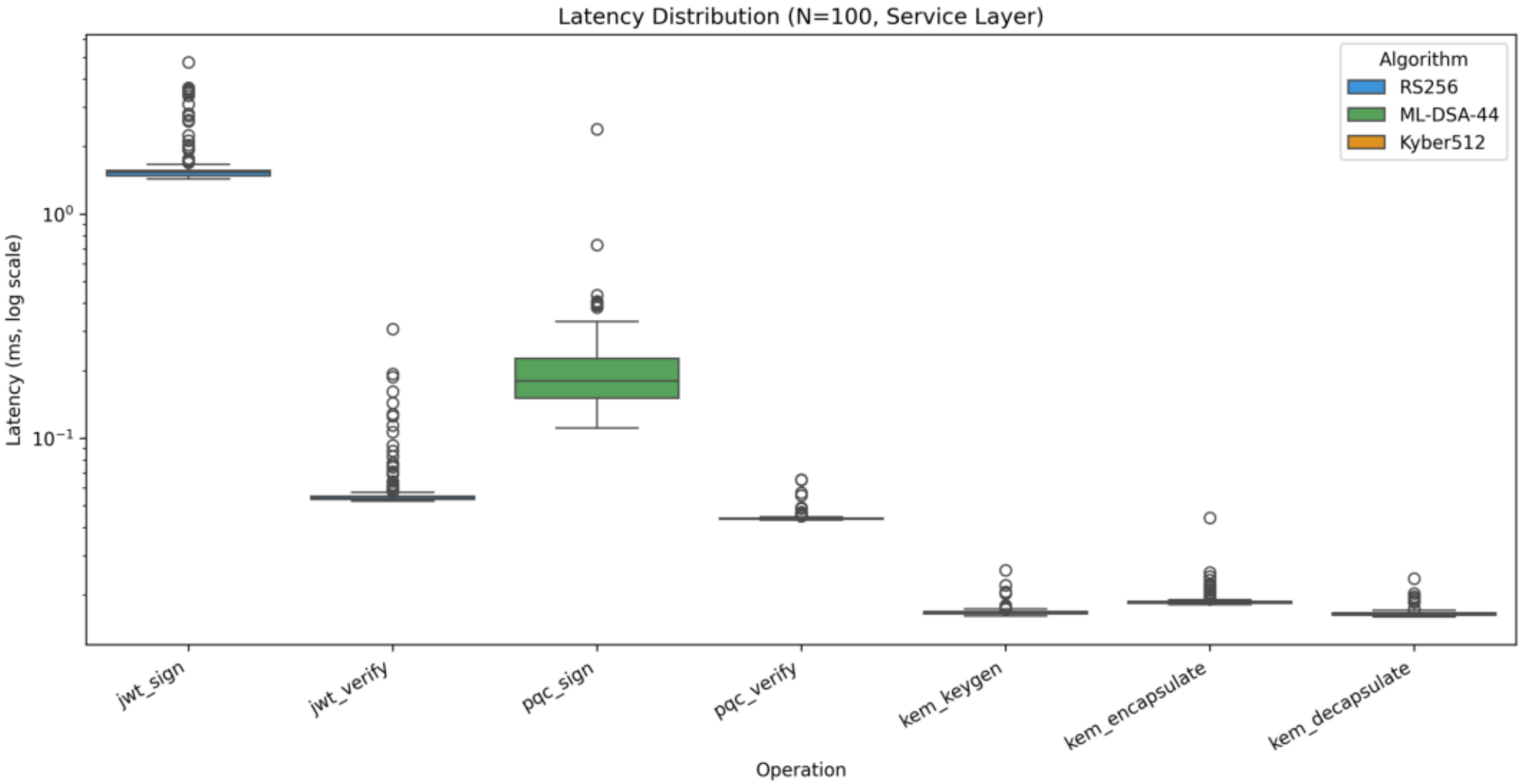
Comparacao Classical vs PQC (Grand Means, 3 runs)

Comparacao	Classical Mean (ms)	PQC Mean (ms)	Speedup
Token Signing	1.6293	0.2031	8.0x
Token Verification	0.0584	0.0442	1.3x
Key Generation (raw)	108.3329	0.0491	2206.4x
Signature (raw)	89.3226	0.1061	841.9x
Verification (raw)	0.1131	0.0441	2.6x

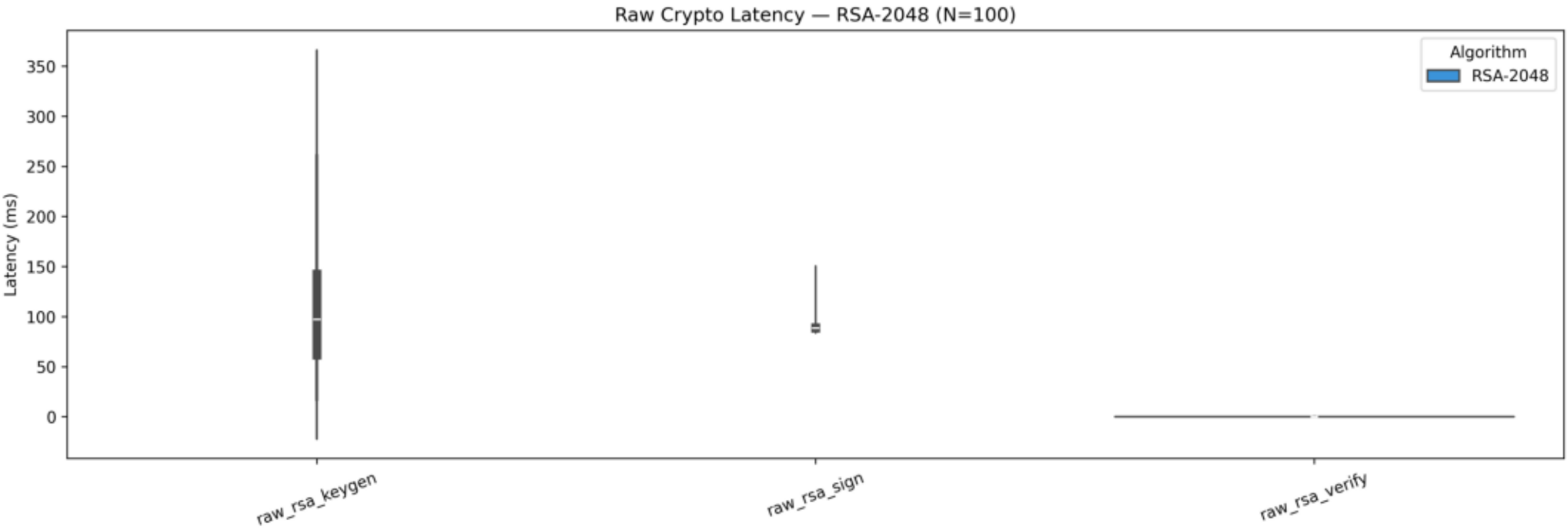
Latencia: Classical (RS256) vs PQC (ML-DSA-44)

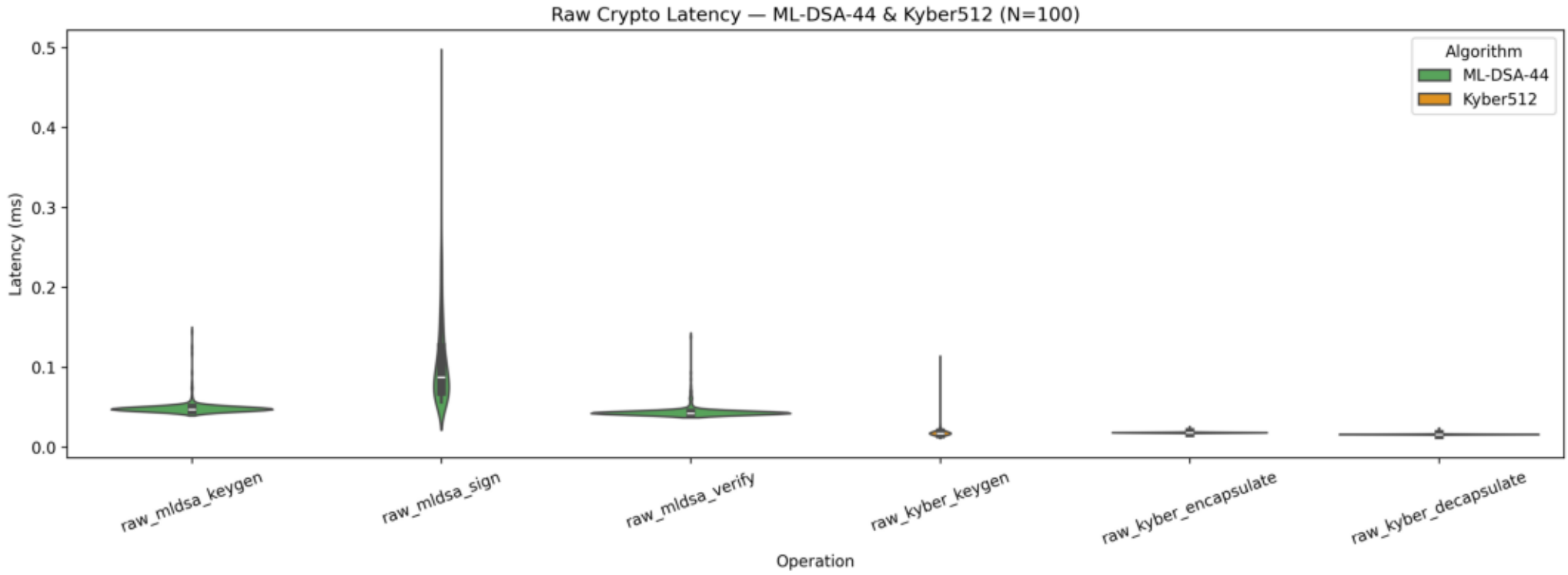


Distribuicao de Latencia (N=300 total, Service Layer)

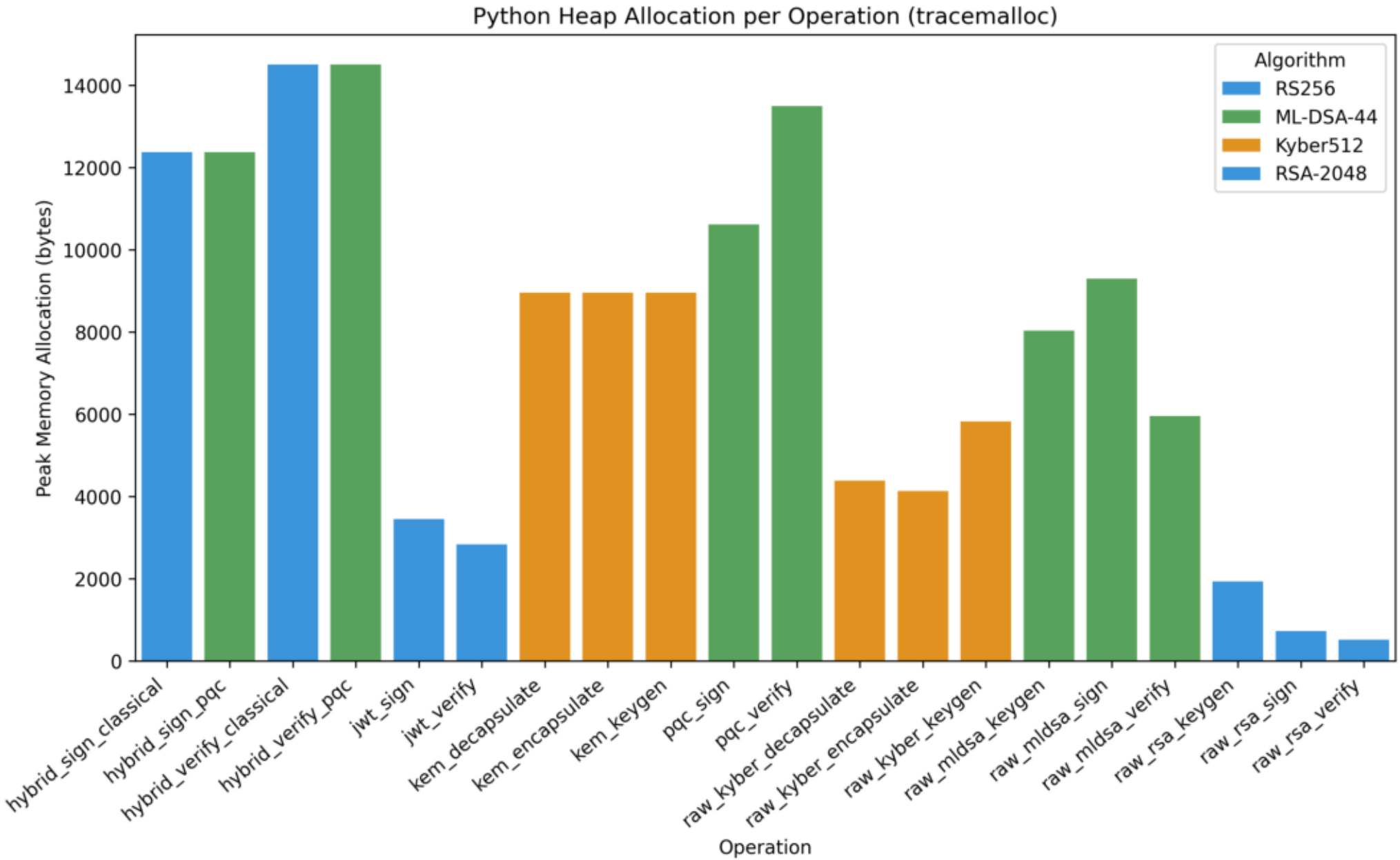


Distribuicao Raw Crypto (Violin Plot)

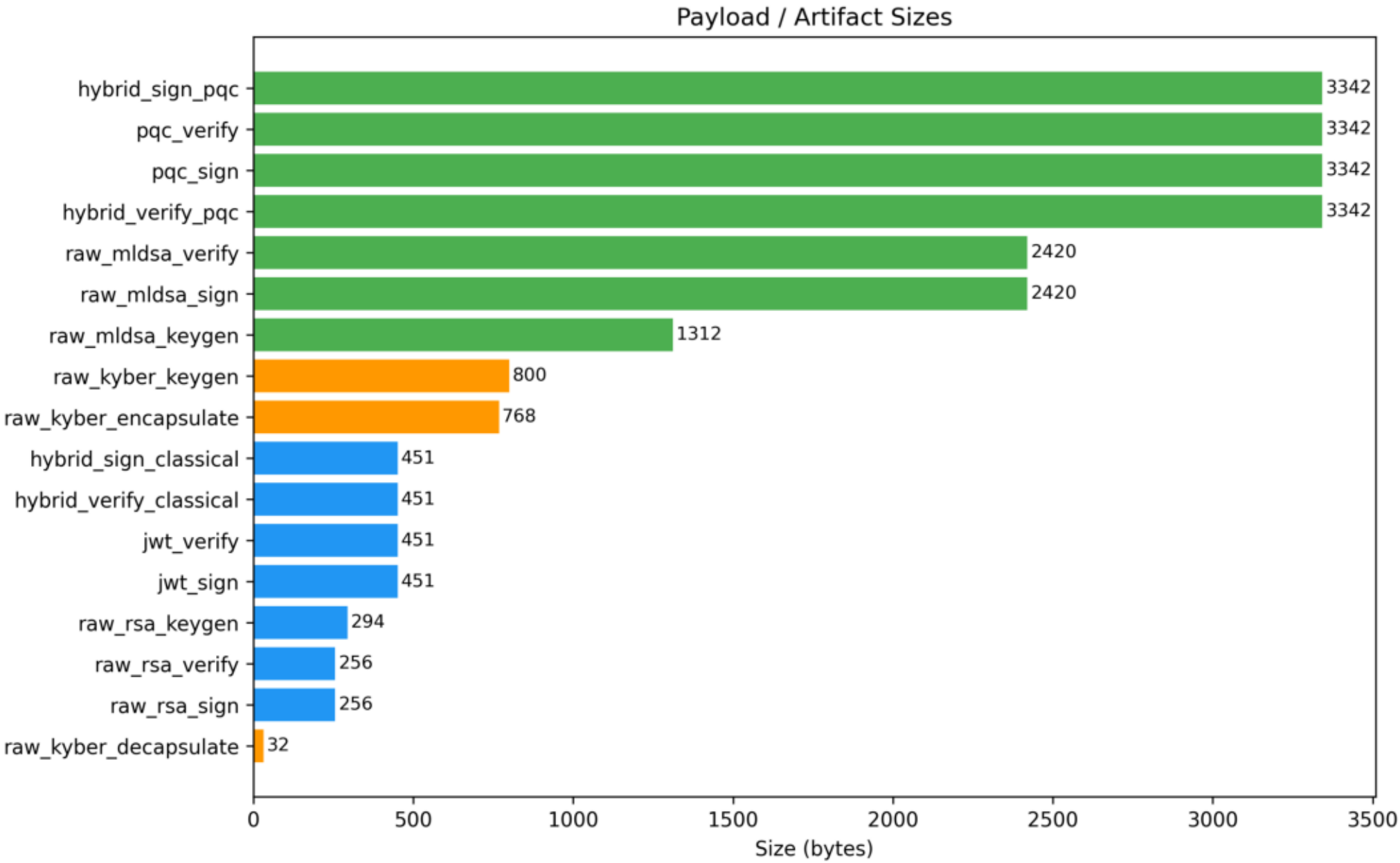




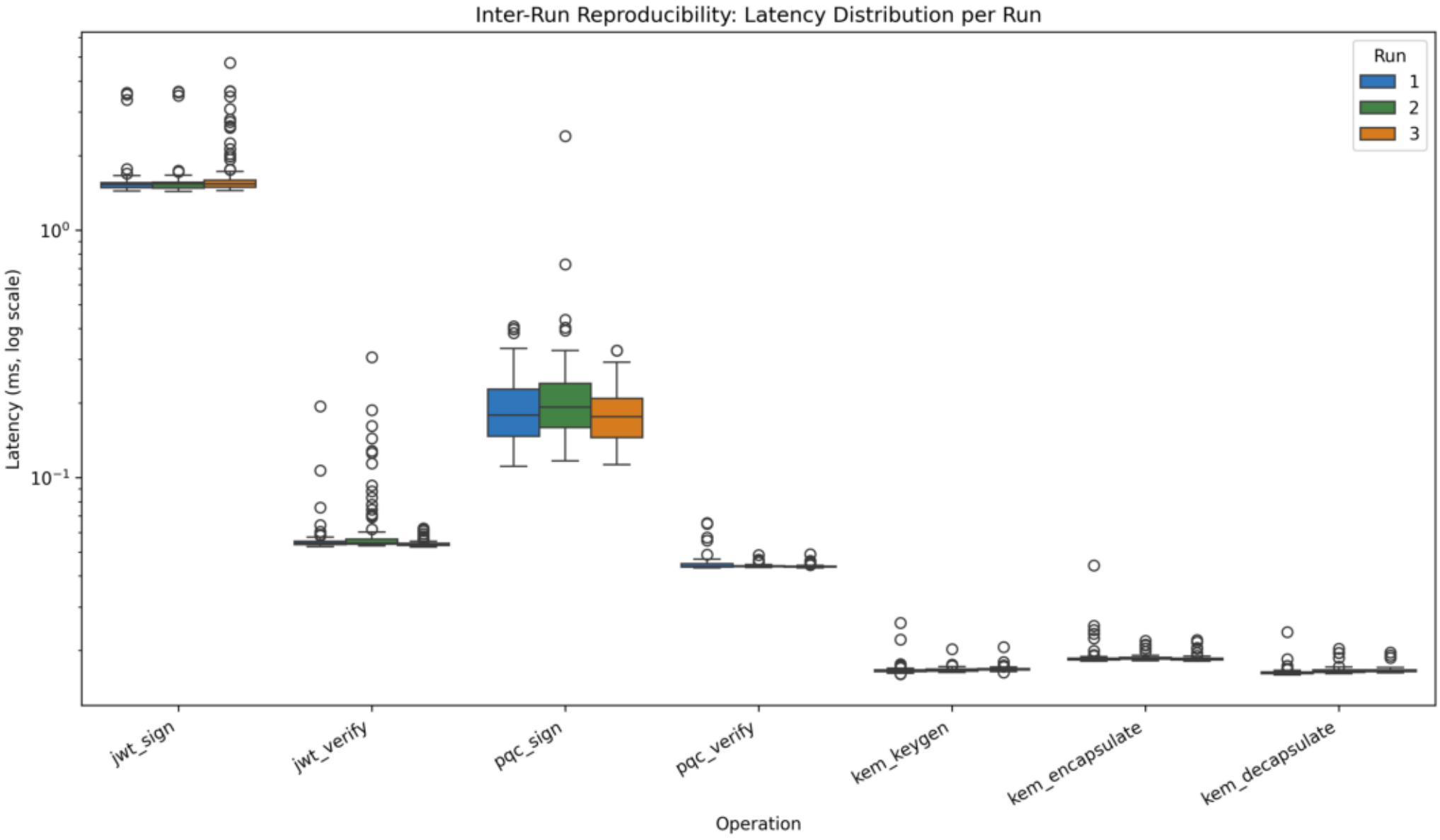
Consumo de Memoria por Operacao (tracemalloc)



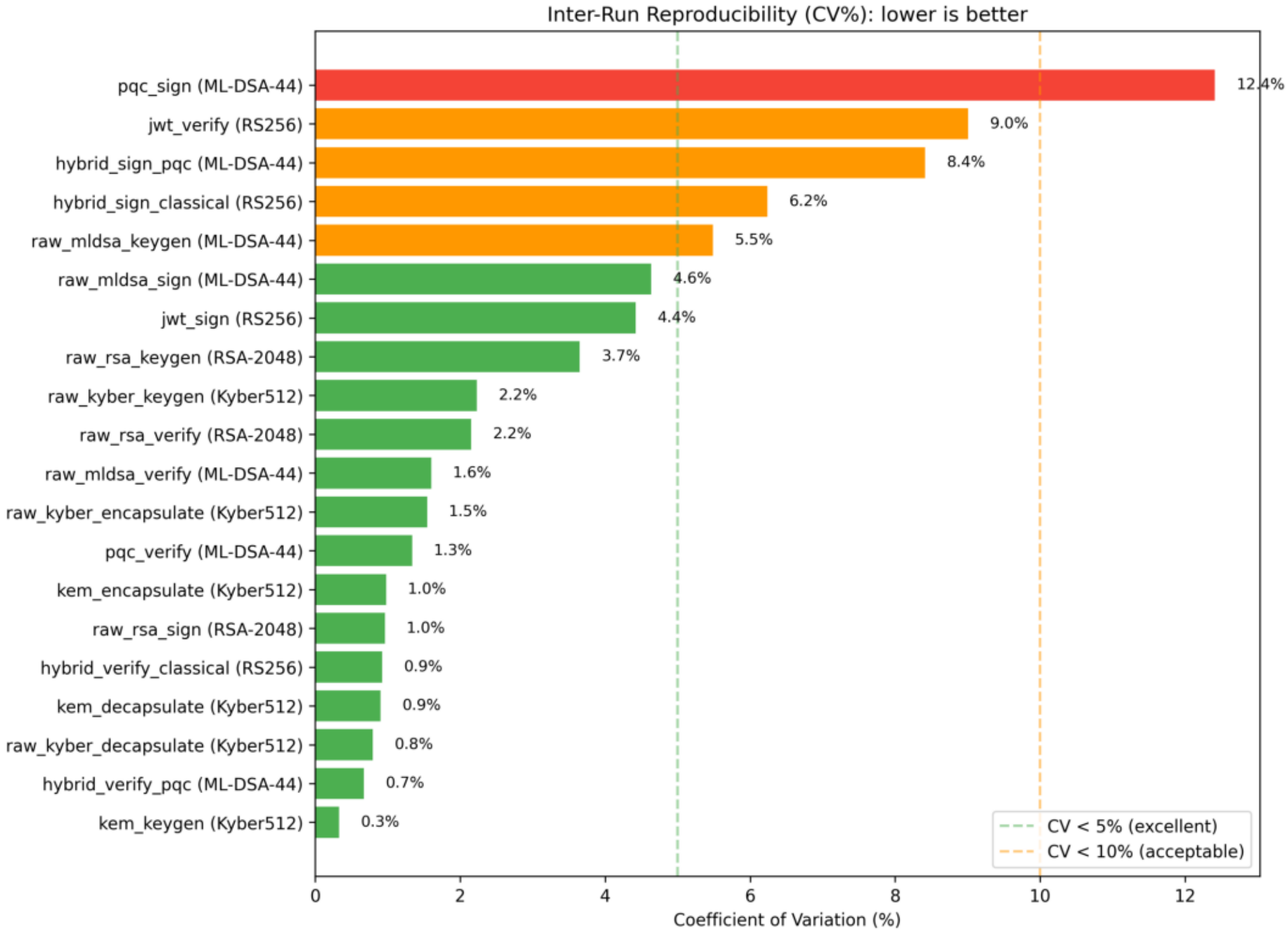
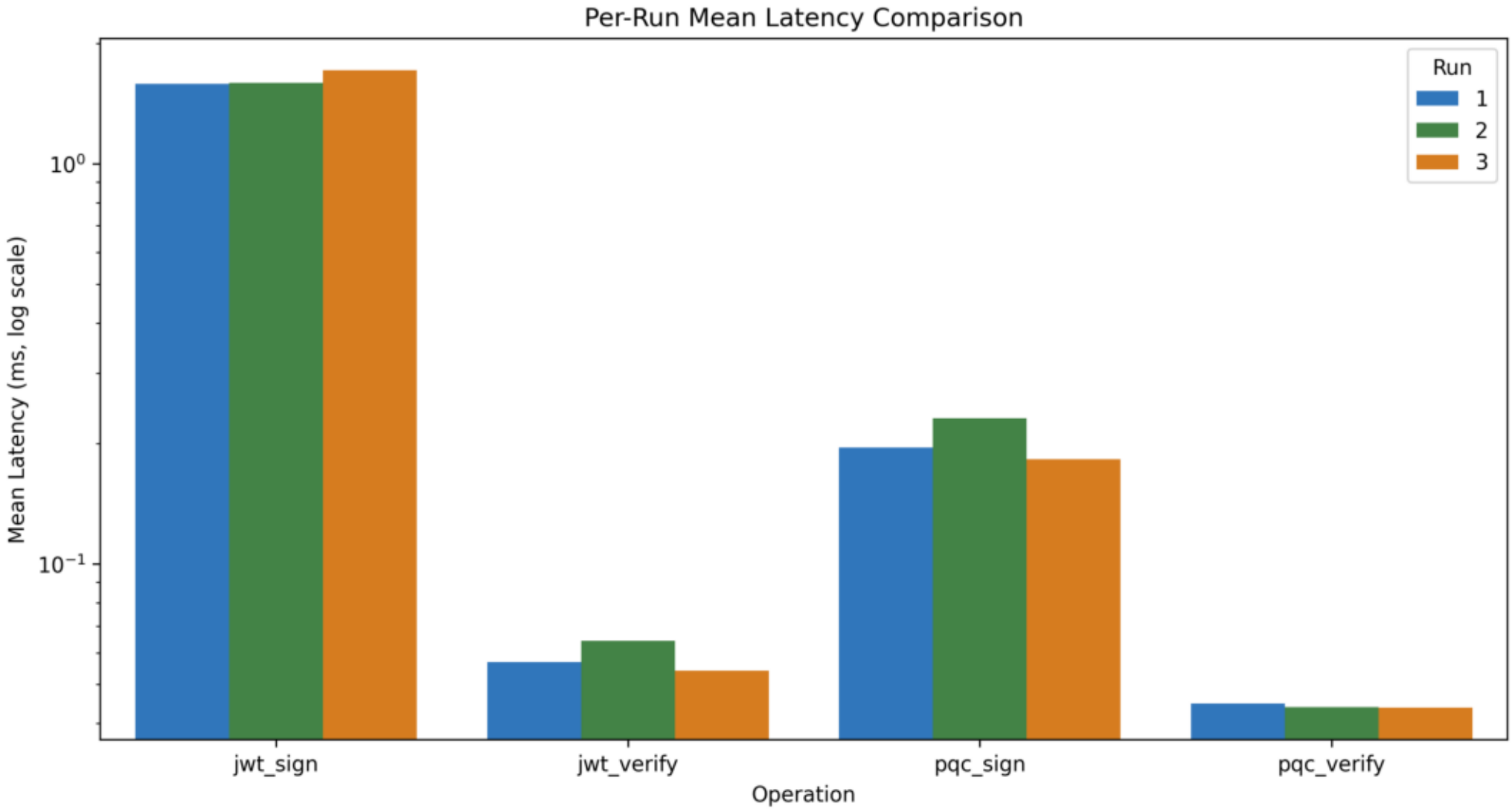
Tamanho de Payloads e Artefatos



Reprodutibilidade: Distribuicao por Run



Coeficiente de Variacao (CV%) por Operacao



Medias por Run — Comparacao Lado a Lado